

The Nature of Software: Software is a logical and tangible product developed to perform specific tasks. Unlike hardware it can't be touched but plays a vital role in controlling and managing the computer operations.

Main properties of software:

1.Intangible: Software isn't something you can touch. It doesn't have a physical form like hardware. It exists as code, instructions and data stored electronically.

Ex: You can't physically hold Google Chrome. It exists only as files and code stored on your device.

2.Prone to "wear out": Software doesn't physically break, but it can become outdated. Bugs, security issues and compatibility problems show up over time. This makes it feel like it has "worn out," even though nothing has physically changed.

Ex: An old version of WhatsApp may stop working on new phones or become insecure over time, even though the app itself didn't "break."

3.Evolves over time: Software is rarely final. It gets updates to fix bugs, improve performance, support new hardware or add new features. This ongoing change is a natural part of its lifecycle.

Ex: Windows 11 receives regular updates that add features, fix bugs and improve security.

4.Custom built: Software is usually developed based on specific requirements. Different users or organizations may need different features, so software is often designed uniquely for its purpose.

Ex: A hospital might get a custom patient-management system developed specifically for its workflows.

5.Requires maintenance: After release, software needs regular updates, debugging and performance improvements. Maintenance is essential to keep it reliable, secure and useful.

Ex: Antivirus software requires frequent updates to detect new viruses.

Software Engineering:

- Software is a set of programs with instructions that help a computer do specific tasks.
- Engineering is about designing and building something useful in an efficient and cost-effective way.

1. Software Engineering combines these ideas. It is the process of planning, creating, testing, and maintaining software.

2. It uses different methods and tools like requirements analysis, design, coding, testing, and maintenance. New technologies keep coming in, so the field grows and changes quickly.

3. By following good practices, developers can create software that is high-quality, reliable, and easy to maintain. Software engineering is mainly used for large and complex software systems, not just small programs.

4. The main goal is to build software that meets user needs while staying within the planned time, budget, and requirements. It ensures the final product is correct, consistent, and delivered on time.

Key Principles of Software Engineering:

1.Modularity: Breaking the software into smaller, reusable components that can be developed and tested independently.

2.Abstraction: Hiding the implementation details of a component and exposing only the necessary functionality to other parts of the software.

3.Encapsulation: Wrapping up the data and functions of an object into a single unit, and protecting the internal state of an object from external modifications.

4.Reusability: Creating components that can be used in multiple projects, which can save time and resources.

5.Maintenance: Regularly updating and improving the software to fix bugs, add new features, and address security vulnerabilities.

6.Testing: Verifying that the software meets its requirements and is free of bugs.

7.Design Patterns: Solving recurring problems in software design by providing templates for solving them.

8.Continuous Integration & Deployment: Continuously integrating the code changes and deploying them into the production environment.

Main Attributes of Software Engineering: Software Engineering is a systematic, disciplined, quantifiable study and approach to the design, development, operation, and maintenance of a software system. There are four main Attributes of Software Engineering:

1.Efficiency: It provides a measure of the resource requirement of a software product efficiently.

2.Reliability: It assures that the product will deliver the same results when used in similar working environment.

3.Reusability: This attribute makes sure that the module can be used in multiple applications.

4.Maintainability: It is the ability of the software to be modified, repaired, or enhanced easily with changing requirements.

Software Process in Software Engineering:

1. Software processes in software engineering refer to the methods and techniques used to develop and maintain software.

2. Some examples of software processes include:

- Waterfall
- Agile
- Scrum
- DevOps

3. Each process has its own set of advantages and disadvantages, and the choice of which one to use depends on the specific project and organization.

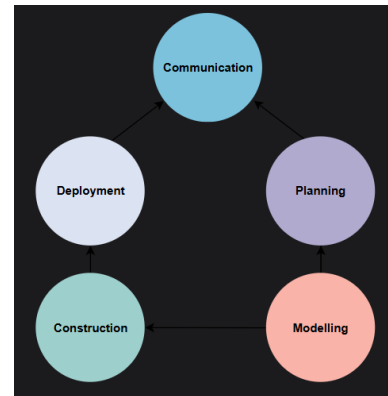
4. The development team needs to choose a software process model and follow it throughout the project. This helps the work stay organized and systematic. Everyone knows what to do next and how to do it. A process model brings clarity and discipline to the whole project. Each phase has clear entry and exit points, so moving from one stage to the next becomes easy and well-defined.

5. Software process provide structured methods for developing and maintaining software. Each has unique strengths, tailored to different project needs. Choosing the right process enhances efficiency and product quality.

Generic process model in software development: The generic process model is an abstraction of the software development process. It is used in most software since it provides a base for them.

The generic process model encompasses the following five steps:

- Communication
- Planning
- Modelling
- Construction
- Deployment



1.Communication: In this step, we communicate with the clients and end-users.

- We discuss the requirements of the project with the users.
- The users give suggestions on the project. If any changes are difficult to implement, we work on alternative ideas.

2.Planning: In this step, we plan the steps for project development. After completing the final discussion, we report on the project.

- Planning plays a key role in the software development process.
- We discuss the risks involved in the project.

3.Modelling: In this step, we create a model to understand the project in the real world. We showcase the model to all the developers. If changes are required, we implement them in this step.

- We develop a practical model to get a better understanding of the project.

4.Construction: In this step, we follow a procedure to develop the final product.

- If any code is required for the project development, we implement it in this phase.
- We also test the project in this phase.

5.Deployment: In this phase, we submit the project to the clients for their feedback and add any missing requirements.

- We get the client feedback.
- Depending on the feedback form, we make the appropriate changes.